Controls HW 4

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September 26, 2026

1 Convexity

a)

Show $f(x) = ||x||^2$ is strictly convex.
Consider 2nd order characterization of convexity which says that a twice differentiable function defined on a convex set with a positive definite hessian is strictly convex

$$\nabla f(x) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}$$

$$\nabla^2 f(x) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I \succ 0 \; \forall x \in \mathbb{R}^2$$

Also $\mathbb{R}^2$ is clearly a convex set since for any 2 points on the set $x, y \in \mathbb{R}^2$ the point $\lambda x + (1-\lambda)y \in \mathbb{R}^2$

b)

i) Show A, B convex $\Rightarrow A \cap B$ convex.
Let $x, y \in A \cap B$. Then $x \in A,\ x \in B$ and $y \in A,\ y \in B$.
A convex $\Rightarrow$ for $x, y \in A$, $z = \lambda x + (1-\lambda)y \in A$
B convex $\Rightarrow$ for $x, y \in B$, $z = \lambda x + (1-\lambda)y \in B$
So $z \in A$ and $z \in B \Rightarrow z \in A \cap B$.

ii) let $V = \left\{ v \in \mathbb{R}^n \text{ s.t. } p_v(x) = \sum_{i=1}^n v_i x^{i-1} \geq 0 \;\; \forall x \in \mathbb{R} \right\}$
Take $x, y \in V$, let $z = \lambda x + (1-\lambda)y$ for $\lambda \in [0, 1]$.

$$x, y \in V \Rightarrow \sum_{i=1}^n x_i c^{i-1} \geq 0, \quad \sum_{i=1}^n y_i c^{i-1} \geq 0$$

$$\sum_{i=1}^n z_i c^{i-1} = \lambda \underbrace{\sum_{i=1}^n x_i c^{i-1}}_{\geq 0} + (1-\lambda) \underbrace{\sum_{i=1}^n y_i c^{i-1}}_{\geq 0}$$

$\lambda, 1-\lambda$ also ≥ 0, so $\sum z_i c^{i-1} \geq 0$, so V is a convex set.

iii) Recall functions $f(x), g(x)$ added: $h(x) = f(x) + g(x)$
Show functions of the form $f(x) = \sum_{i=0}^{N} a_i x^i \geq 0$ is convex.
Let both

$$g_1(x) = \sum_{i=0}^{N_1} a_i x^i \geq 0, \qquad g_2(x) = \sum_{i=0}^{N_2} b_i x^i \geq 0$$

Assume w.l.o.g. that $N_1 > N_2$. Then let $h(x) = \lambda g_1(x) + (1-\lambda) g_2(x)$:

$$h(x) = \underbrace{\lambda}_{\geq 0} \underbrace{\sum_{i=0}^{N_1} a_i x^i}_{\geq 0} + \underbrace{(1-\lambda)}_{\geq 0} \underbrace{\sum_{i=0}^{N_2} b_i x^i}_{\geq 0} \;\Rightarrow\; h(x) \geq 0$$

$$h(x) = \sum_{i=0}^{N_2} \left(\lambda a_i x^i + (1-\lambda) b_i x^i\right) + \sum_{j=N_2+1}^{N_1} \lambda a_j x^j$$

$$= \sum_{i=0}^{N_2} \underbrace{\left(\lambda a_i + (1-\lambda) b_i\right)}_{c_i} x^i + \sum_{j=N_2+1}^{N_1} \underbrace{\lambda a_j}_{c_j} x^j = \sum_{i=0}^{N_1} c_i x^i \geq 0$$

Problem 2: Quadratic Programming

Find the closest pair of points $a, b \in \mathbb{R}^2$
s.t. a satisfies $y = -2x + 5$ (i.e., (a_1, a_2) satisfies this)
and s.t. b on unit box:

$$0 \leq b_1 \leq 1, \qquad 0 \leq b_2 \leq 2$$

Write as a QP optimization problem:

$$\min_z \; \tfrac{1}{2} z^T H z + z^T f, \qquad z = (a_1, a_2, b_1, b_2)^T$$

$$\text{s.t.} \quad Az \leq b, \qquad A_{eq} z = b_{eq}$$

and returns z^*, solves for a^*, b^*.

Inequality constraints

a_1, a_2: no constraints. $b_1 \leq 1$, $-b_1 \leq 0$, $b_2 \leq 2$, $-b_2 \leq 0$:

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \qquad Az \leq \begin{pmatrix} 1 \\ 0 \\ 2 \\ 0 \end{pmatrix}$$

Equality constraints

Recall: $a_2 = -2a_1 + 5 \implies -2a_1 + a_2 = 5$

$$A_{eq} = \begin{pmatrix} -2 & 1 & 0 & 0 \end{pmatrix}, \qquad b_{eq} = (5)$$

$$A_{eq} z = b \;\Rightarrow\; (-2,\ 1,\ 0,\ 0) \begin{pmatrix} a_1 \\ a_2 \\ b_1 \\ b_2 \end{pmatrix} = (5)$$

Objective

We want a, b as close as possible, i.e.

$$\min \|a-b\|^2 = (a-b)^T(a-b) = a^Ta - a^Tb - b^Ta + b^Tb = a^Ta - 2a^Tb + b^Tb$$

Let

$$\begin{pmatrix} a & b \end{pmatrix} (H) \begin{pmatrix} a \\ b \end{pmatrix}, \qquad H = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix}$$

$$\begin{pmatrix} a & b \end{pmatrix}^T \begin{pmatrix} H_{11}a + H_{12}b \\ H_{21}a + H_{22}b \end{pmatrix} = a^TH_{11}a + a^TH_{12}b + b^TH_{21}a + b^TH_{22}b$$

So the final quadratic form:

$$a^Ta - 2a^Tb + b^Tb \;\Rightarrow\; H_{11} = H_{22} = I_2, \quad H_{12} = H_{21} = -I_2$$

and it matches up. No need for the z^Tf term, so let $f = \vec{0}$. So we have

$$H = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}$$

So final quadratic:

$$\min_{z=(a,b)} \; z^THz$$

subject to

$$(-2,\ 1,\ 0,\ 0)\, z = (5)$$

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \end{pmatrix} z \le \begin{pmatrix} 1 \\ 0 \\ 2 \\ 0 \end{pmatrix}$$

3) Parametric Programming

Dynamics:

$$x_{k+1} = x_k + u_k, \qquad u_k, x_k \in \mathbb{R}^2 \qquad \Longrightarrow \qquad x_n = x_0 + \sum_{i=0}^{n-1} u_i.$$

Let $\mathbf{1}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, so $\mathbf{1}_2^\top = \begin{pmatrix} 1 & 1 \end{pmatrix}$. Consider the parametric program

$$J(x_0) = \min_{u_0,\dots,u_9} \quad \frac{1}{2} \sum_{i=0}^{9} u_i^\top u_i \tag{1}$$

$$\text{s.t.} \quad \mathbf{1}_2^\top x_{10} = \mathbf{1}_2^\top \Big(x_0 + \sum_{i=0}^{9} u_i \Big) \le 1. \tag{2}$$

This finds the minimum-effort inputs $\{u_i\}$ (given initial state x_0) such that $\mathbf{1}_2^\top x_{10} \le 1$.

(a)

Write the KKT conditions and use them to show $u_i^* = u_j^*$ for all $i, j \in \{0, \dots, 9\}$.

Stationarity (general form):

$$\nabla f + \nabla h^\top \lambda + \nabla g^\top \mu = 0,$$

where the equality-constraint term $\nabla h^\top \lambda$ drops out (no equality constraints).

Rewrite with the stacked decision variable

$$z = \begin{pmatrix} u_0 \\ \vdots \\ u_9 \end{pmatrix} \in \mathbb{R}^{20}, \qquad f(z) = \frac{1}{2} \sum_{i=0}^{9} u_i^\top u_i, \qquad g(z) = \mathbf{1}_2^\top \Big(x_0 + \sum_{i=0}^{9} u_i \Big) - 1 \le 0.$$

Then

$$\nabla_z f = \begin{pmatrix} u_0 \\ \vdots \\ u_9 \end{pmatrix}, \qquad \nabla_z g = \begin{pmatrix} \mathbf{1}_2 \\ \vdots \\ \mathbf{1}_2 \end{pmatrix} = \mathbf{1}_{20}.$$

So by stationarity ($\sum_i \mu_i \nabla g_i$, where i indexes the constraints; here there is one, so $\mu \in \mathbb{R}$):

$$\nabla_z f(z) + \mu \nabla_z g(z) = \begin{pmatrix} u_0 \\ \vdots \\ u_9 \end{pmatrix} + \mu \begin{pmatrix} \mathbf{1}_2 \\ \vdots \\ \mathbf{1}_2 \end{pmatrix} = 0 \quad \Longrightarrow \quad u_i^* = -\mu \, \mathbf{1}_2 \ \ \forall i.$$

$$\Longrightarrow \quad u_0^* = \cdots = u_9^* = u^* = -\mu \, \mathbf{1}_2.$$

The remaining KKT conditions are primal feasibility $g(z) \le 0$, dual feasibility $\mu \ge 0$, and complementary slackness $\mu \, g(z) = 0$.

(b) Find the optimal input $u^*(x_0)$ and cost $J^*(x_0)$ when the constraint is active.

Active constraint:

$$\mathbf{1}_2^\top \Big(x_0 + \sum_{i=0}^{9} u_i \Big) = 1.$$

Using the same variable z as before, active $\Rightarrow$ equality constraint ($\mu \geq 0$):

$$\begin{aligned} L(z,\mu) &= f(z) + \mu\, g(z) \\ &= \frac{1}{2} \sum_{i=0}^{9} u_i^\top u_i + \mu \Big(\mathbf{1}_2^\top \Big(x_0 + \sum_{i=0}^{9} u_i \Big) - 1 \Big) \end{aligned}$$

$$\nabla_z L = \begin{pmatrix} u^* \\ \vdots \\ u^* \end{pmatrix} + \mu \begin{pmatrix} \mathbf{1}_2 \\ \vdots \\ \mathbf{1}_2 \end{pmatrix} = 0 \quad \Longrightarrow \quad u^* = -\mu\, \mathbf{1}_2$$

$$\nabla_\mu L = \mathbf{1}_2^\top \Big(x_0 + \sum_{i=0}^{9} u_i \Big) - 1 = 0.$$

Let $x_0 = \begin{pmatrix} x_{0,1} \\ x_{0,2} \end{pmatrix}$. Substituting $u_i = u^* = -\mu\, \mathbf{1}_2$:

$$\mathbf{1}_2^\top \big(x_0 - 10\mu\, \mathbf{1}_2 \big) = x_{0,1} + x_{0,2} - 20\mu = 1.$$

So

$$\mu = \frac{\mathbf{1}_2^\top x_0 - 1}{20} \quad \Longrightarrow \quad u^*(x_0) = -\mu\, \mathbf{1}_2 = \frac{1 - \mathbf{1}_2^\top x_0}{20}\, \mathbf{1}_2.$$

The region of x_0 where the constraint is active is simply where

$$\mathbf{1}_2^\top x_0 \geq 1.$$

In this region $\mu = \frac{\mathbf{1}_2^\top x_0 - 1}{20} \geq 0$, so dual feasibility and primal feasibility remain satisfied.

$$J^*(x_0) = \frac{1}{2} \sum_{i=0}^{9} {u^*}^\top u^* = \frac{1}{2} \cdot 10 \cdot 2 \left(\frac{1 - \mathbf{1}_2^\top x_0}{20} \right)^2 = \frac{\big(1 - \mathbf{1}_2^\top x_0\big)^2}{40}.$$

(c) Find the optimal input $u^*(x_0)$ and cost $J^*(x_0)$ when the constraint is not active.

Inactive constraint:

$$\mathbf{1}_2^\top \Big(x_0 + \sum_{i=0}^{9} u_i \Big) < 1 \quad \Longrightarrow \quad \mu = 0.$$

Stationarity:

$$\nabla_z f = \begin{pmatrix} u^* \\ \vdots \\ u^* \end{pmatrix} = 0 \quad \Longrightarrow \quad u^* = 0.$$

$$J^*(x_0) = \frac{1}{2}\sum_{i=0}^{9} 0^\top 0 = 0.$$

The constraint is inactive when

$$\mathbf{1}_2^\top x_0 < 1.$$

At the border, $\mathbf{1}_2^\top x_0 = 1 \;\Rightarrow\; u^* = \dfrac{1-1}{20}\mathbf{1}_2 = 0$ and $J^* = 0$ ✓ (the two solutions agree).

4) Min Snap Quadrotor Trajectories

Recall that the (planar) quadrotor is differentially flat, so we can control any trajectory in the flat outputs:

$$\gamma(t) = \begin{pmatrix} y(t) \\ z(t) \end{pmatrix}.$$

Define $n+1$ waypoints for the trajectory, $w_0, \ldots, w_n \in \mathbb{R}^2$. Represent the path γ_i between w_i and w_{i+1} as a polynomial:

$$\gamma_i(t_i) = \sum_{j=0}^{d-1} \sigma_{i,j}\, t_i^j, \qquad t_i \in [0, \Delta t_i],$$

where $\sigma_{i,j} \in \mathbb{R}^2$ are the polynomial coefficients, chosen to be feasible and efficient.

Decision variable:

$$v = \big(\sigma_{0,0}, \ldots, \sigma_{0,d-1}, \ldots, \sigma_{n-1,d-1}\big).$$

Goal: formulate min-snap trajectory optimization as a QP. (Min snap $\Longleftrightarrow$ min control effort for quadrotors.)

(a) Constraints in terms of σ, Δt, w, d

(i) The i-th polynomial must start at w_i and end at w_{i+1}, i.e. $\gamma_i(0) = w_i$ and $\gamma_i(\Delta t_i) = w_{i+1}$. So

$$\sum_{j=0}^{d-1} \sigma_{i,j}\, t_i^j \Big|_{t_i=0} = w_i \quad \Longrightarrow \quad \sigma_{i,0} = w_i,$$

and

$$\sum_{j=0}^{d-1} \sigma_{i,j}\, t_i^j \Big|_{t_i=\Delta t_i} = w_{i+1} \quad \Longrightarrow \quad \sum_{j=0}^{d-1} \sigma_{i,j}\, (\Delta t_i)^j = w_{i+1}.$$

(ii) The k-th derivative of γ_i ($k \in \{1, \ldots, 4\}$) must equal the k-th derivative of γ_{i+1} where they meet:

$$\gamma_i^{(k)}(\Delta t_i) = \gamma_{i+1}^{(k)}(0).$$

$k = 1$:

$$\gamma_i^{(1)}(t_i) = 0 + \sum_{j=1}^{d-1} j\, \sigma_{i,j}\, t_i^{j-1}, \qquad \text{so} \qquad \sum_{j=1}^{d-1} j\, \sigma_{i,j}\, (\Delta t_i)^{j-1} = (1)\, \sigma_{i+1,1}.$$

$k = 2$:

$$\sum_{j=2}^{d-1} j(j-1)\, \sigma_{i,j}\, (\Delta t_i)^{j-2} = (2)\, \sigma_{i+1,2}.$$

$k = 3$:

$$\sum_{j=3}^{d-1} j(j-1)(j-2)\, \sigma_{i,j}\, (\Delta t_i)^{j-3} = (6)\, \sigma_{i+1,3}.$$

$k = 4$:

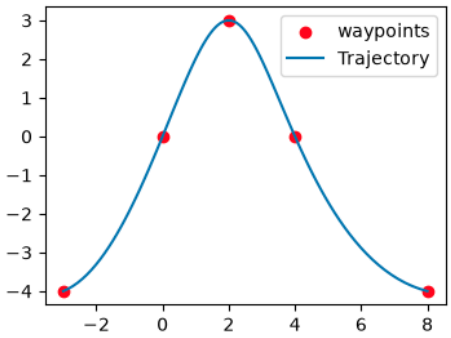
$$\sum_{j=4}^{d-1} j(j-1)(j-2)(j-3)\, \sigma_{i,j}\, (\Delta t_i)^{j-4} = 24\, \sigma_{i+1,4}.$$

In general, for $k \in \{1, \ldots, 4\}$ and $i \in \{0, \ldots, n-2\}$:

$$\sum_{j=k}^{d-1} \frac{j!}{(j-k)!}\, \sigma_{i,j}\, (\Delta t_i)^{j-k} = k!\, \sigma_{i+1,k}.$$

b)

Wrote down both equality constraints and derivative constraints in code as matrices (can check code: `hw4/pos_constraints.py`). Then when we run our min snap optimization problem with drake solver we get



c) Minimizing snap as objective

Approximate effort as the integral of snap squared:

$$\min_z J(z) = \sum_{i=0}^{n-1} \int_0^{\Delta t_i} \left\| \frac{d^4 \gamma_i}{dt_i^4} \right\|^2 dt_i$$

(i) Evaluate the integral

$$\frac{d^4 \gamma_i}{dt^4} = \sum_{j=4}^{d-1} \frac{j!}{(j-4)!} \sigma_{i,j} (\Delta t_i)^{j-4}, \qquad \text{let } c_j = \frac{j!}{(j-4)!}$$

$$\left\| \frac{d^4 \gamma_i}{dt^4} \right\|^2 = \gamma_i^{(4)T} \gamma_i^{(4)} = \sum_{j=4}^{d-1} \sum_{l=4}^{d-1} c_j c_l (\Delta t_i)^{j-4} (\Delta t_i)^{l-4} \sigma_{i,j}^T \sigma_{i,l} = \sum_{j=4}^{d-1} \sum_{l=4}^{d-1} c_j c_l \, t_i^{j+l-8} \, \sigma_{i,j}^T \sigma_{i,l}$$

So

$$\int_0^{\Delta t_i} \|\gamma_i^{(4)}\|^2 \, dt_i = \sum_{j=4}^{d-1} \sum_{l=4}^{d-1} c_j c_l \, \sigma_{i,j}^T \sigma_{i,l} \int_0^{\Delta t_i} t_i^{j+l-8} \, dt_i = \sum_{j=4}^{d-1} \sum_{l=4}^{d-1} c_j c_l \, \frac{(\Delta t_i)^{j+l-7}}{j+l-7} \, \sigma_{i,j}^T \sigma_{i,l}$$

General quadratic form:

$$s^T Q s = \sum_i \sum_j Q_{ij} \, s_i s_j$$

Split the dot product by axis:

$$\sigma_{i,j}^T \sigma_{i,l} = \sigma_{i,j}^{(y)} \sigma_{i,l}^{(y)} + \sigma_{i,j}^{(z)} \sigma_{i,l}^{(z)}$$

So

$$\int \|\gamma_i^{(4)}\|^2 \, dt_i = \sum_j \sum_l c_j c_l \frac{(\Delta t_i)^{j+l-7}}{j+l-7} \left(\sigma_{i,j}^{(y)} \sigma_{i,l}^{(y)} \right) + \sum_j \sum_l c_j c_l \frac{(\Delta t_i)^{j+l-7}}{j+l-7} \left(\sigma_{i,j}^{(z)} \sigma_{i,l}^{(z)} \right)$$

Let $Q^{(i)}$ be defined by

$$Q_{j,l}^{(i)} = \begin{cases} c_j c_l \dfrac{(\Delta t_i)^{j+l-7}}{j+l-7} & j, l \geq 4 \\ 0 & \text{else} \end{cases}$$

so

$$\int \|\gamma_i^{(4)}\|^2 = s_y^T Q^{(i)} s_y + s_z^T Q^{(i)} s_z$$

where

$$s_y = \left(\sigma_{0,0}^y, \ldots, \sigma_{n-1,d-1}^y \right)^T, \qquad s_z = \left(\sigma_{0,0}^z, \ldots, \sigma_{n-1,d-1}^z \right)^T$$

d)

(i)

Write trajectory optimization problem

$$\min_v \sum_{i=0}^{n-1} \int_0^{\Delta t_i} \|\gamma_i^{(4)}(t_i)\|^2 \, dt_i$$

subject to

$$A_i v = b_i, \qquad A_{i,2k} v = b_{i,2k} \quad \text{for } i = 0, \ldots, n$$

As

$$\min_v \tfrac{1}{2} v^T H v \qquad \text{s.t. } A_{eq} v = b_{eq}$$

Cost derivation (matrix form):

$$\sum_{i=0}^{n-1} \int \|\gamma_i^{(4)}\|^2 \, dt_i \;\rightarrow\; \sum_{i=0}^{n-1} \left(\underbrace{\sum_{j=4}^{d-1} \sum_{l=4}^{d-1} c_j c_l \frac{\Delta t_i^{\,j+l-7}}{j+l-7} \, \sigma_{i,j}^T \sigma_{i,l}}_{\text{Inner Sum}} \right)$$

Let the constant term inside the sum be $C_{j,l}$ (capital).
Here

$$v_i = (\sigma_{i,0}, \ldots, \sigma_{i,d-1})^T, \qquad \sigma_{i,j} \in \mathbb{R}^2$$

so elements of $Q_i \in \mathbb{R}^{2\times 2}$, not scalars, since for each element (j,l):

$$\sigma_{i,j}^T (Q_i)_{j,l} \, \sigma_{i,l} = C_{j,l} \, \sigma_{i,j}^T \sigma_{i,l} \;\Rightarrow\; (Q_i)_{j,l} = C_{j,l} \, I_2$$

So let $C \in \mathbb{R}^{d\times d}$, with

$$(C)_{j,l} = \begin{cases} C_{j,l} & j,l \geq 4 \\ 0 & \text{else} \end{cases}$$

$$Q_i = C \otimes I_2 \quad \text{(sends each element of } C \text{ to elt} \cdot I_2)$$

Now consider the total cost:

$$J(v) = \sum_{i=0}^{n-1} v_i^T Q_i v_i, \qquad v = \begin{pmatrix} v_0 \\ \vdots \\ v_n \end{pmatrix}$$

As a full matrix:

$$(v_0, \ldots, v_n) \begin{pmatrix} Q_0 & 0 & \cdots & 0 \\ 0 & Q_1 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & Q_N \end{pmatrix} \begin{pmatrix} v_0 \\ \vdots \\ v_n \end{pmatrix} \;\Rightarrow\; H = I_n \otimes Q_i$$

(assuming Δt_i constant $\forall i$; any $i \in \{0, \dots, n-1\}$ works)
So now we have

$$J(v) = \sum_{i=0}^{n-1} \|\gamma_i^{(4)}\|^2 = v^T H v \qquad \text{(quadratic cost)}$$

Constraints: for the i-th spline γ_i:
Equality:

$$\sigma_{i,0} = \omega_i \in \mathbb{R}^2, \qquad \sum_{j=0}^{d-1} \sigma_{i,j} (\Delta t_i)^j = \omega_{i+1}$$

Derivative: for $k = 1, \dots, 4$:

$$\sum_{j=k}^{d-1} c_j (\Delta t_i)^{j-k} \sigma_{i,j} - k! \, \sigma_{i+1,k} = 0$$

Let $v_i = (\sigma_{i,0}, \dots, \sigma_{i,d-1})$, $v = (v_0, \dots, v_{n-1})$.

Equality constraint (A_{pos})

In terms of v_i:

$$a_{pos,i} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \Delta t_i^0 & \Delta t_i^1 & \cdots & \Delta t_i^{d-1} \end{pmatrix}$$

Since this multiplies v_i, a vector of $\sigma_{i,j} \in \mathbb{R}^2$:

$$A_{pos,i} = \begin{pmatrix} I_2 & 0_2 & \cdots & 0_2 \\ I_2 \Delta t_i^0 & I_2 \Delta t_i^1 & \cdots & I_2 \Delta t_i^{d-1} \end{pmatrix} \in \mathbb{R}^{4 \times (2d)}$$

i.e., using the same trick of letting each scalar element $\to I_2$ (multiply block):

$$A_{pos,i} = a_{pos,i} \otimes I_2$$

So for $i = 0, \dots, n-1$:

$$A_{pos,i} \, v_i = \begin{pmatrix} I_2 & \cdots & 0_2 \\ \Delta t_i I_2 & \cdots & \Delta t_i^{d-1} I_2 \end{pmatrix} \begin{pmatrix} \sigma_{i,0} \\ \vdots \\ \sigma_{i,d-1} \end{pmatrix} = \begin{pmatrix} \omega_i \\ \omega_{i+1} \end{pmatrix}$$

So for $v = (v_0, \dots, v_{n-1})^T$: $A_{pos,i} \in \mathbb{R}^{4 \times 2d}$, $b_{pos,i} \in \mathbb{R}^4$.
Let $A_{pos} = \text{blockdiag}(A_{pos,i}) \in \mathbb{R}^{4n \times 2nd}$:

$$A_{pos} = \begin{pmatrix} A_{pos,0} & & 0 \\ & \ddots & \\ 0 & & A_{pos,n-1} \end{pmatrix} \begin{pmatrix} v_0 \\ \vdots \\ v_{n-1} \end{pmatrix} = \begin{pmatrix} b_{pos,0} \\ \vdots \\ b_{pos,n-1} \end{pmatrix}$$

Derivative constraint

Considering $\sigma_{i,j}$ as scalars, let $a_{der,i,k} \in \mathbb{R}^{1\times(nd)}$ be a row vector:

$$a_{der,i,k} = \big(\underbrace{0,\dots,0,\, p_{i,k}}_{\text{corresponds to } v_i},\quad \underbrace{-q_k,\, 0,\dots,0}_{\text{corresponds to } v_{i+1}}\big)$$

where

$$p_{i,k} = \big(\underbrace{0,\dots,0}_{\text{first } k \text{ terms are } 0},\quad \underbrace{c_j(\Delta t_i)^{j-k}}_{\text{for } j=k,\dots,d-1}\big)$$

and $-q_k = (0,0,\dots,-k!,\dots,0)$, with $-k!$ in the k-th element.

To make $a_{der,i,k}$ work with $\sigma_{i,j} \in \mathbb{R}^2$:

$$a_{der,i,k} = a_{der,i,k} \otimes I_2 \in \mathbb{R}^{2\times(2nd)}$$

Since this works with $v \in \mathbb{R}^{2nd}$, stack into the full derivative constraint matrix:

$$A_{der} = \begin{pmatrix} a_{der,0,0} \\ \vdots \\ a_{der,0,4} \\ \vdots \\ a_{der,n-1,4} \end{pmatrix}, \qquad A_{der}\begin{pmatrix} v_0 \\ \vdots \\ v_{n-1} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$A_{der} \in \mathbb{R}^{8n\times 2nd}, \qquad b_{der} \in \mathbb{R}^{8n}$$

Final constraint matrix:

$$A = \begin{pmatrix} A_{pos} \\ A_{der} \end{pmatrix} \in \mathbb{R}^{12n\times 2nd}, \qquad v \in \mathbb{R}^{2nd}, \qquad b = \begin{pmatrix} b_{pos} \\ b_{der} \end{pmatrix} \in \mathbb{R}^{12n}$$

Problem 5

Consider the optimization problem, $x \in \mathbb{R}^n$:

$$\min_x f(x) \quad \text{s.t.} \quad g(x) \le 0,\ g(x) \in \mathbb{R}^n$$

Lagrangian:

$$L(x,\lambda) = f(x) + \lambda^T g(x), \qquad \lambda \ge 0$$

Define the dual function as the unconstrained min problem:

$$q(\lambda) = \min_x L(x,\lambda)$$

(a)

Consider feasible x s.t. $g(x) \leq 0$, and feasible λ s.t. $\lambda \geq 0$. Show:

$$q(\lambda) \leq L(x, \lambda) \leq f(x)$$

Since $q(\lambda) = \min_x L(x, \lambda) \implies q(\lambda) \leq L(x, \lambda)\ \forall x$.

$$L(x, \lambda) = f(x) + \lambda^T g(x)$$

Since $\lambda \geq 0$ and $g(x) \leq 0$:

$$\lambda^T g(x) \leq 0 \implies f(x) \geq f(x) + \lambda^T g(x) = L(x, \lambda)$$

(b)

Consider $f(x) = \frac{1}{2} x^T Q x$, $g(x) = Ax - b$, $Q \succ 0$ symmetric.

$$\min_x \tfrac{1}{2} x^T Q x \quad \text{s.t. } Ax \leq b \quad \xrightarrow{\text{dual}} \quad q(\lambda) = \min_x \tfrac{1}{2} x^T Q x + \lambda^T (Ax - b) = \min_x \tfrac{1}{2} x^T Q x + \lambda^T A x - \lambda^T b$$

(c)

$$\max_\lambda q(\lambda) \quad \text{s.t. } \lambda \geq 0 \quad \iff \quad \min_\lambda -q(\lambda) \quad \text{s.t. } \lambda \geq 0$$

Solve the inner minimization over x:

$$\nabla_x L(x, \lambda) = Qx + A^T \lambda = 0 \implies Qx = -A^T \lambda \implies x^* = -Q^{-1} A^T \lambda$$

Substitute back:

$$q(\lambda) = \tfrac{1}{2} \left(-Q^{-1} A^T \lambda\right)^T Q \left(-Q^{-1} A^T \lambda\right) + \lambda^T A \left(-Q^{-1} A^T \lambda\right) - \lambda^T b$$

$$= \tfrac{1}{2} \lambda^T A Q^{-1} A^T \lambda - \lambda^T A Q^{-1} A^T \lambda - \lambda^T b = -\tfrac{1}{2} \lambda^T (A Q^{-1} A^T) \lambda - \lambda^T b$$

So

$$q(\lambda) = -\tfrac{1}{2} \lambda^T (A Q^{-1} A^T) \lambda - \lambda^T b, \qquad -q(\lambda) = \tfrac{1}{2} \lambda^T (A Q^{-1} A^T) \lambda + \lambda^T b$$

Since $Q \succ 0 \implies Q^{-1} \succ 0 \implies A Q^{-1} A^T \succeq 0$.
So $\min_\lambda -q(\lambda)$ is convex, over the convex set $\lambda \geq 0$.

(d)

Let x^*, λ^* satisfy KKT. Show $q(\lambda^*) = f(x^*)$.

KKT conditions:

1) $\nabla f + \nabla g^T \lambda = 0 \implies Q x^* + A^T \lambda^* = 0$

2) $g(x^*) \leq 0$

3) $\lambda^* \geq 0$

4) $\lambda_i^* g_i(x^*) = 0$ (complementary slackness)

From stationarity in λ:

$$\nabla_\lambda(-q(\lambda)) = AQ^{-1}A^T\lambda + b = 0 \implies \lambda^* = -(AQ^{-1}A^T)^{-1}b$$

Then

$$q(\lambda^*) = L(x^*, \lambda^*) = \tfrac{1}{2}x^{*T}Qx^* + \lambda^{*T}(Ax^* - b)$$

By complementary slackness, $\lambda^{*T}(Ax^* - b) = 0$, so

$$q(\lambda^*) = \tfrac{1}{2}x^{*T}Qx^* = f(x^*)$$